Practical-2

Problem Definition :

Design and implement a 4-bit digital calculator that performs basic arithmetic operations on two 4-bit operands A (A3..A0) and B CO2, CO3/PO1 (B3..B0). The calculator should support the following operations selected by a 2-bit opcode Op1 Op0:

● 00 — Addition: S = A + B (4-bit result + carry out)

● 01 — Subtraction: S = A − B using two's complement (detect overflow/borrow)
● 10 — Multiplication: P = A × B (produce an 8-bit product)
● 11 — Optional / Extended: pass-through / increment A / unsigned divide (choose one for extension)

1. Implement at gate-level (use standard ICs such as full adders, XOR, AND/OR/MUX) and test in TinkerCad Circuits (or equivalent simulator).
2. Provide truth table(s), block diagram, logic derivations, test vectors and final circuit schematic.

Tools/Technology To Be Used :

● TinkerCad Circuits — build gate-level design using logic primitives (AND, OR, XOR, NOT) and ICs (74xx family) or breadboard components in the simulator.

● Optionally use Logisim (or similar) for easier logic visualisation.

● Use a multimeter/logic probe or LEDs to show outputs or 7 segment display modules for human-readable results.

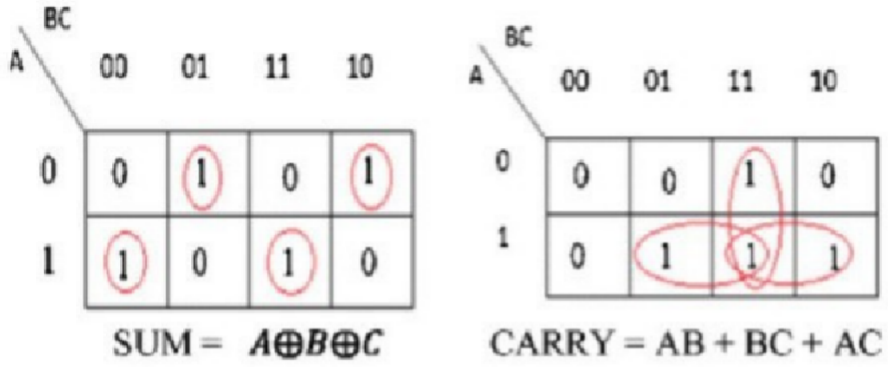
1) Full Adder:

Description: A Full Adder is a combinational logic circuit that performs the arithmetic addition of three binary bits. Unlike a half adder, it can handle a carry bit from a previous stage, making it the fundamental building block for multi-bit binary addition.

Truth Table:

A	B	C_{in}	Sum (S)	C_{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

K-Map:



Equation:

1. Sum Equation

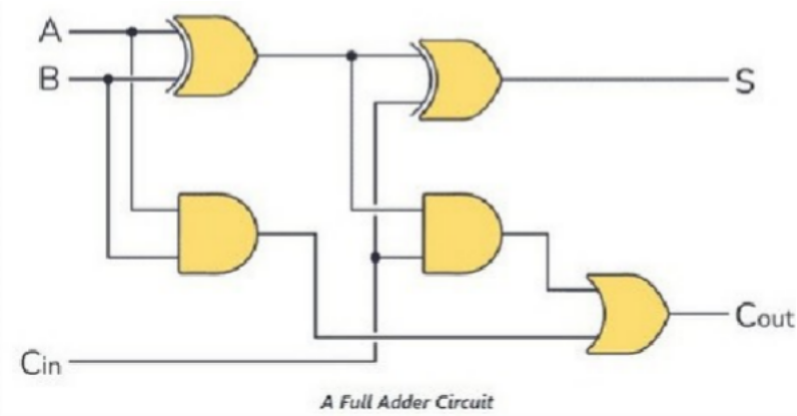
The Sum output is obtained using XOR operation:

$S=A \oplus B \oplus Cin$

2. Carry Equation

The Carry output is $Cout = AB + BCin + ACin$

Circuit:



A Full Adder Circuit

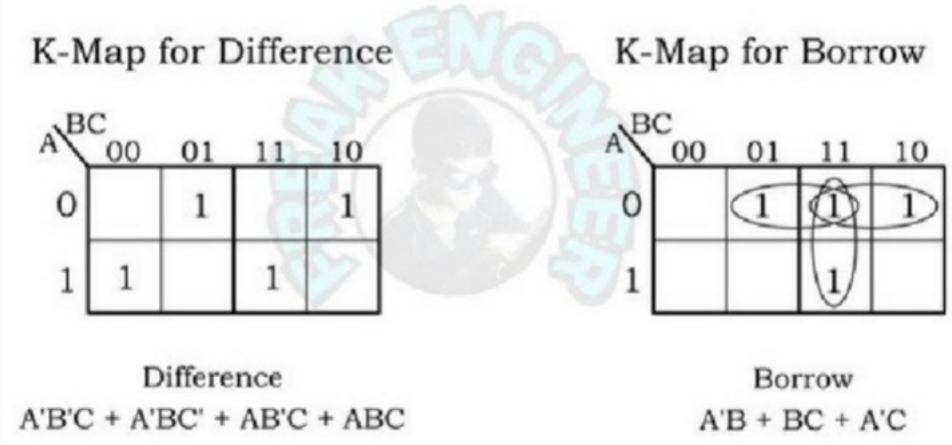
2) Full Subtractor:

Description: A Full Subtractor is a combinational logic circuit that performs the subtraction of three binary bits: the minuend, the subtrahend, and a borrow-in from a previous lower-order bit position. Unlike a Half Subtractor, which can only handle two bits, a full subtractor is essential for multi-bit subtraction as it accounts for the borrow generated during the subtraction of less significant bits.

Truth table:

A	B	Bin	Difference (D)	Borrow-out (Bout)
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

K-Map:

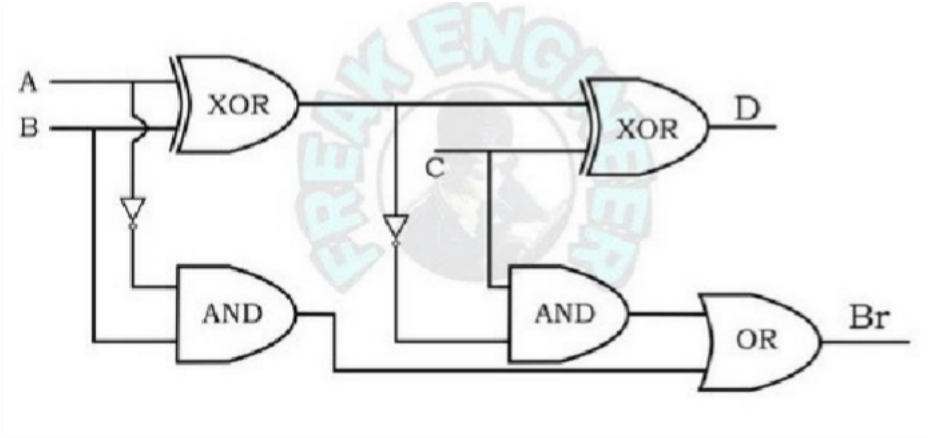


Equation:

Difference : $D=A \oplus B \oplus Bin$

Borrow: $Bo = u\ 't\ AB + ABin + {}'BBin$

Circuit:



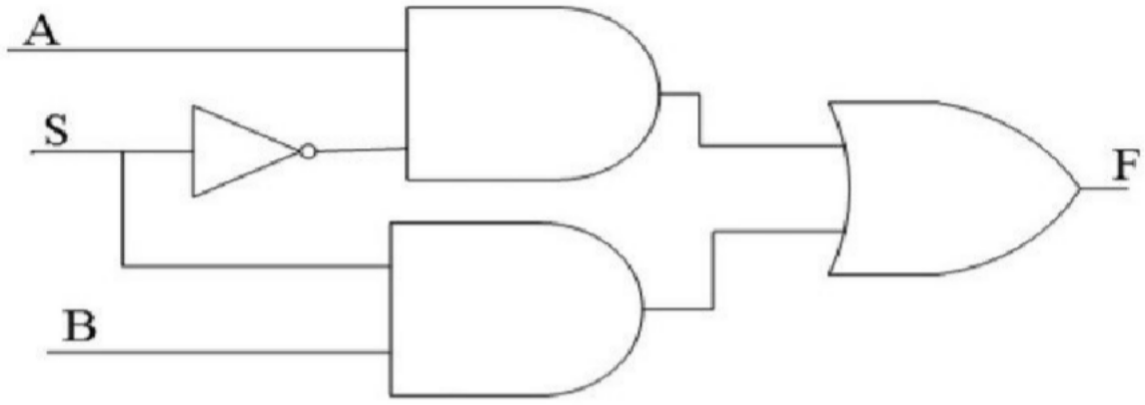
3) Multiplexer:

Description:A multiplexer (commonly abbreviated as MUX) is a combinational logic circuit that selects one of several digital or analog input signals and forwards the selected input into a single output line. Often called a data selector, it acts like a multi-position electronic switch that determines which input reaches the output based on a set of control inputs known as select lines.

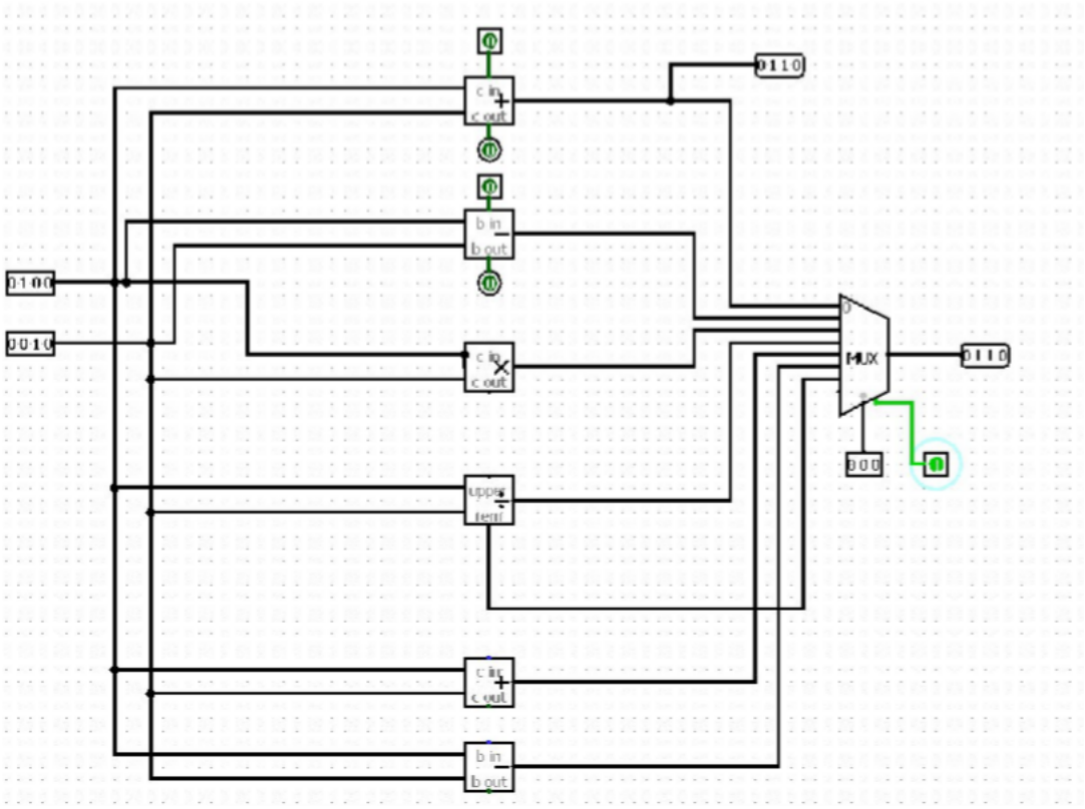
Equation:

$Y=I0S'+I1S$

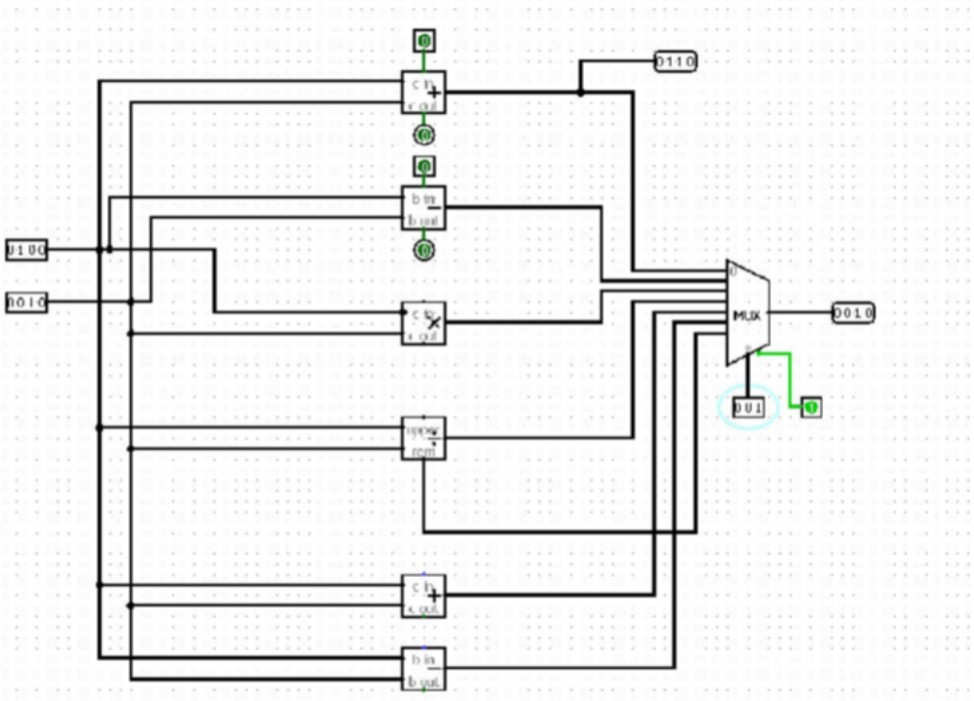
Circuit:



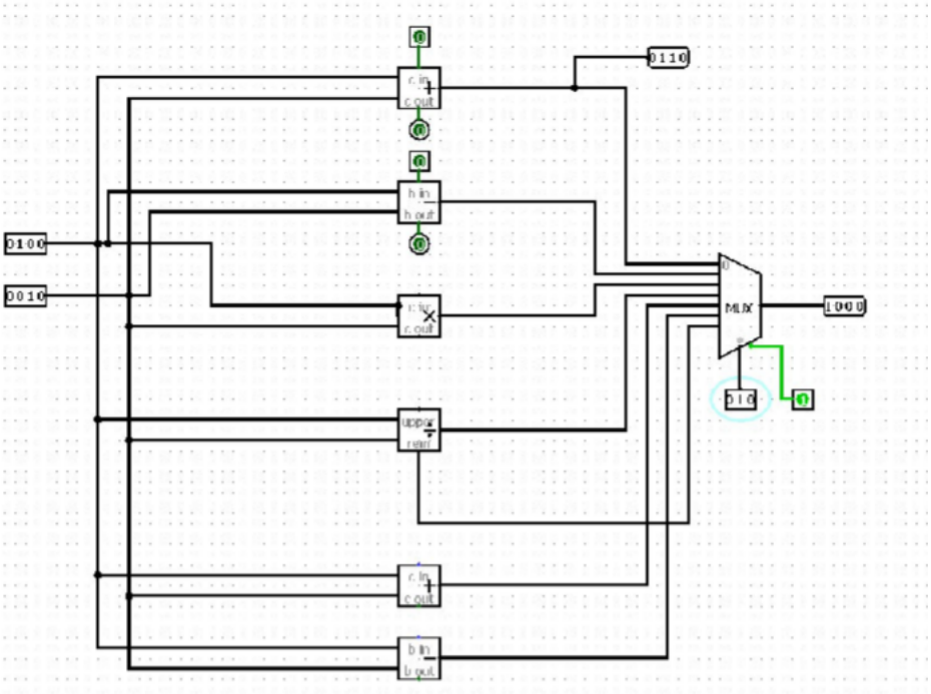
Addition:



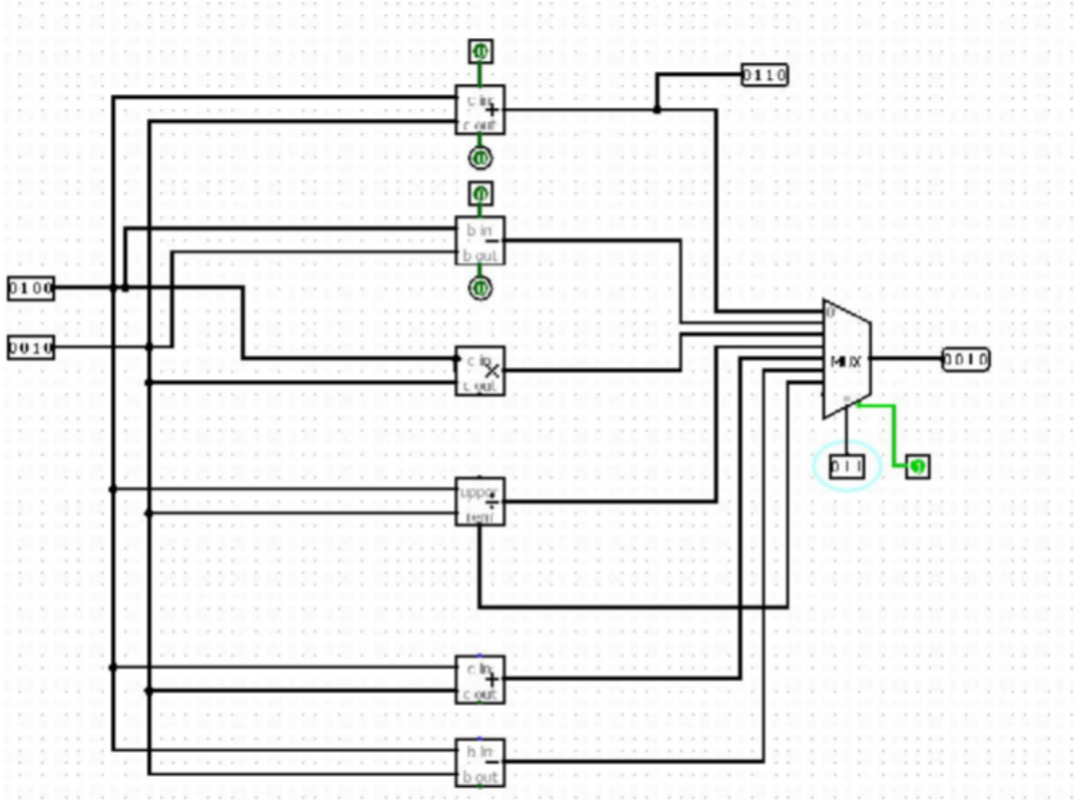
Subtraction:



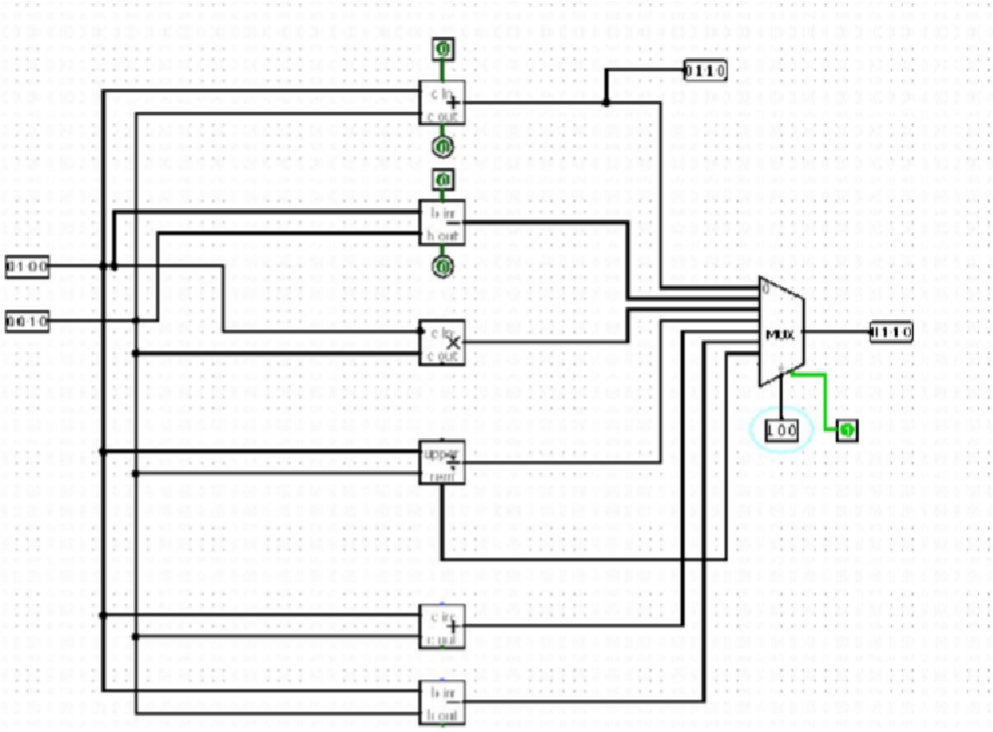
Multiply:



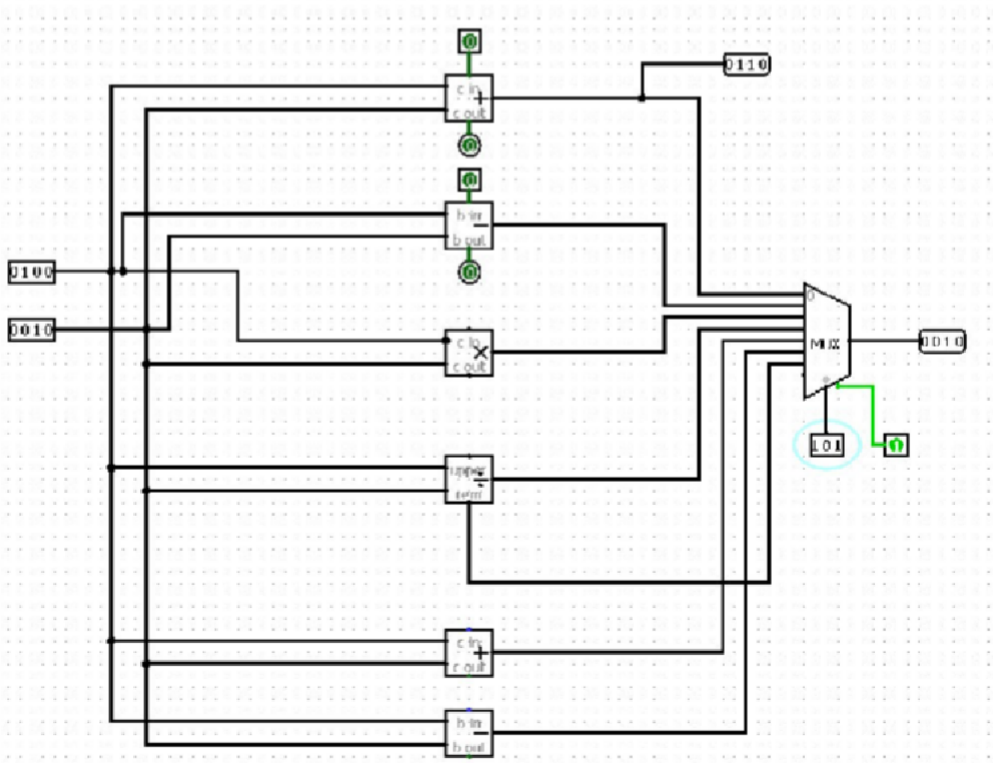
Division:



Increment:



Decrement:



Conclusion: This practical session on the 4-bit digital calculator illustrates the core principles of ALU (Arithmetic Logic Unit) design. By combining standard components like Full Adders, XOR gates for subtraction logic, and AND-gate arrays for multiplication, the circuit successfully processes multi-bit data based on specific control signals. The simulation confirmed that 2-bit opcodes can effectively multiplex diverse functions—ranging from basic addition to 8-bit multiplication—into a single output bus. Ultimately, the project validates the efficiency of binary arithmetic and the scalability of gate-level logic in building functional computational hardware.